

Assignment 10: Data Scraping

Yimei Du

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
#Install familiar packages
library(tidyverse);library(lubridate);library(viridis);library(here)
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
#install.packages("rvest")
library(rvest)

#install.packages("dataRetrieval")
library(dataRetrieval)

#install.packages("tidycensus")
library(tidycensus)

# Set theme
mytheme <- theme_gray() +
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
```

```
theme_set(mytheme)

here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the URL to be scraped. (In other words, read the contents into an **rvest** webpage object.)

```
#2
URL <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024')
class(URL)
```

```
## [1] "xml_document" "xml_node"
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - PWSID
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3
name <- URL %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()
name
```

```
## [1] "Durham"
```

```
id <- URL %>%
  html_nodes("td tr:nth-child(1) td:nth-child(5)") %>%
  html_text()
id
```

```
## [1] "03-32-010"
```

```
ownership <- URL %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()
ownership
```

```
## [1] "Municipality"
```

```
MGD <- URL %>%
  html_nodes("th~ td+ td, th~ td+ td") %>%
  html_text()
MGD
```

```
## [1] "34.5000" "36.0600" "37.3300" "32.1000" "46.6500" "37.3600" "38.2000"
## [8] "41.9000" "36.5800" "36.7300" "42.9600" "34.4500"
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

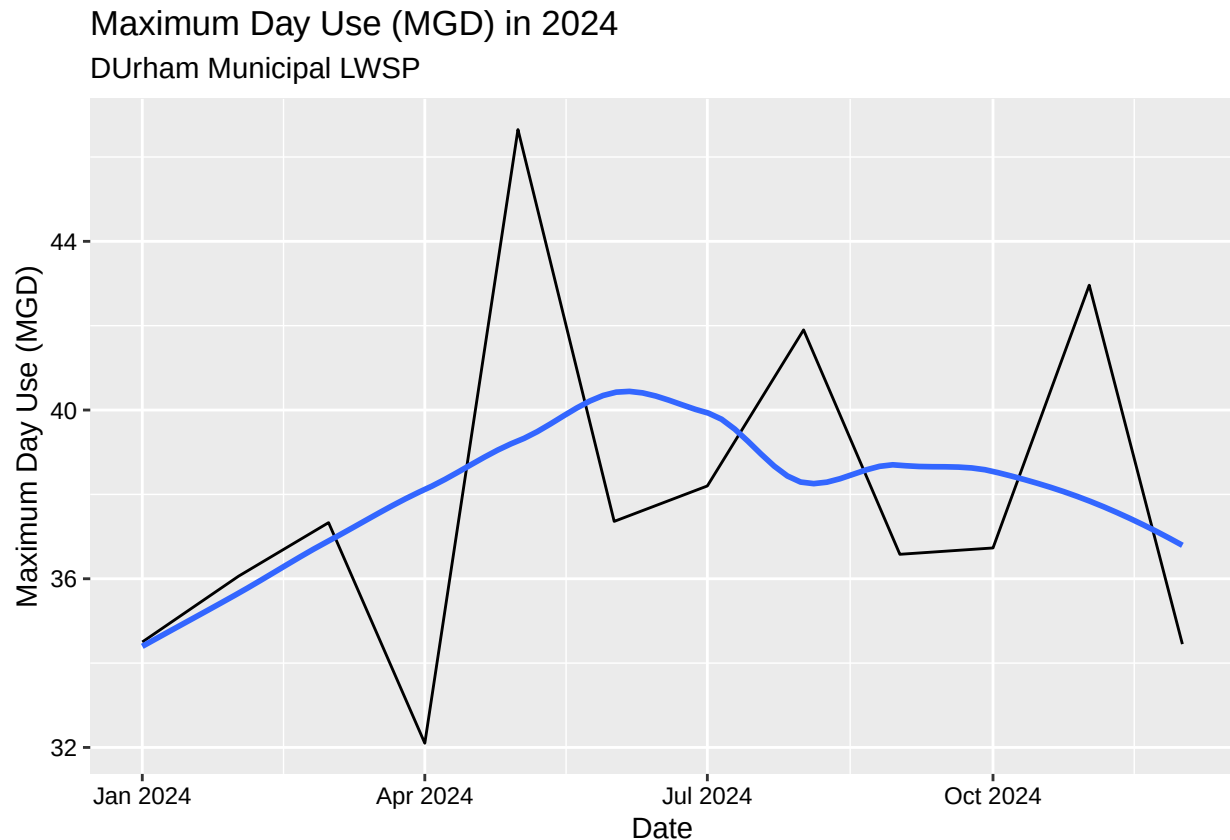
```
#4
df_durham <- data.frame("Month" =
  c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sept", "Oct", "Nov", "Dec"),
  "Year" = rep(2024, 12),
  "Maximum_Day_Use" = as.numeric(MGD))

df_durham <- df_durham %>%
  mutate(
    Date = my(paste(Month, Year))) %>%
  arrange(Date)

#5
ggplot(df_durham, aes(x=Date, y=Maximum_Day_Use)) +
```

```
geom_line() +
geom_smooth(method="loess",se=FALSE) +
labs(title = paste("Maximum Day Use (MGD) in 2024"),
      subtitle = "DURham Municipal LWSP",
      y="Maximum Day Use (MGD)",
      x="Date")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - “PWSID” and “year” - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6.
scrape.it <- function(year, id){

the_durham_url <- read_html(
  paste0('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=', id, '&year=', year)
)
```

```

name <- 'div+ table tr:nth-child(1) td:nth-child(2)'
id <- 'td tr:nth-child(1) td:nth-child(5)'
ownership <- 'div+ table tr:nth-child(2) td:nth-child(4)'
MGD <- 'th~ td+ td, th~ td+ td'

name <- the_durham_url %>% html_nodes(name) %>% html_text()
id <- the_durham_url %>% html_nodes(id) %>% html_text()
ownership <- the_durham_url %>% html_nodes(ownership) %>% html_text()
MGD <- the_durham_url %>% html_nodes(MGD) %>% html_text()

df_url <- data.frame("Month" =
                      c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sept", "Oct", "Nov", "Dec"),
                      "Year" = rep(year, 12),
                      "Maximum_Day_Use" = as.numeric(MGD),
                      "Water_System" = rep(name, 12),
                      "PWSID" = rep(id, 12),
                      "Ownership" = rep(ownership, 12)) %>%
  mutate(
    Date = my(paste(Month, Year))) %>%
  arrange(Date)
  return(df_url)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7
df_2020 <- scrape.it(2020, "03-32-010")
str(df_2020)

```

```

## 'data.frame':    12 obs. of  7 variables:
## $ Month          : chr  "Jan" "Feb" "Mar" "Apr" ...
## $ Year            : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ Maximum_Day_Use: num  36 37 41.7 32 40.6 ...
## $ Water_System    : chr  "Durham" "Durham" "Durham" "Durham" ...
## $ PWSID           : chr  "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
## $ Ownership       : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Date            : Date, format: "2020-01-01" "2020-02-01" ...

```

```

ggplot(df_2020, aes(x=Date, y=Maximum_Day_Use)) +
  geom_line() +
  geom_smooth(method="loess", se=FALSE) +
  labs(
    title = paste(
      "Maximum Day Use (MGD) in 2020"
    ),
    y="Maximum Day Use",
    x="Date"
  )

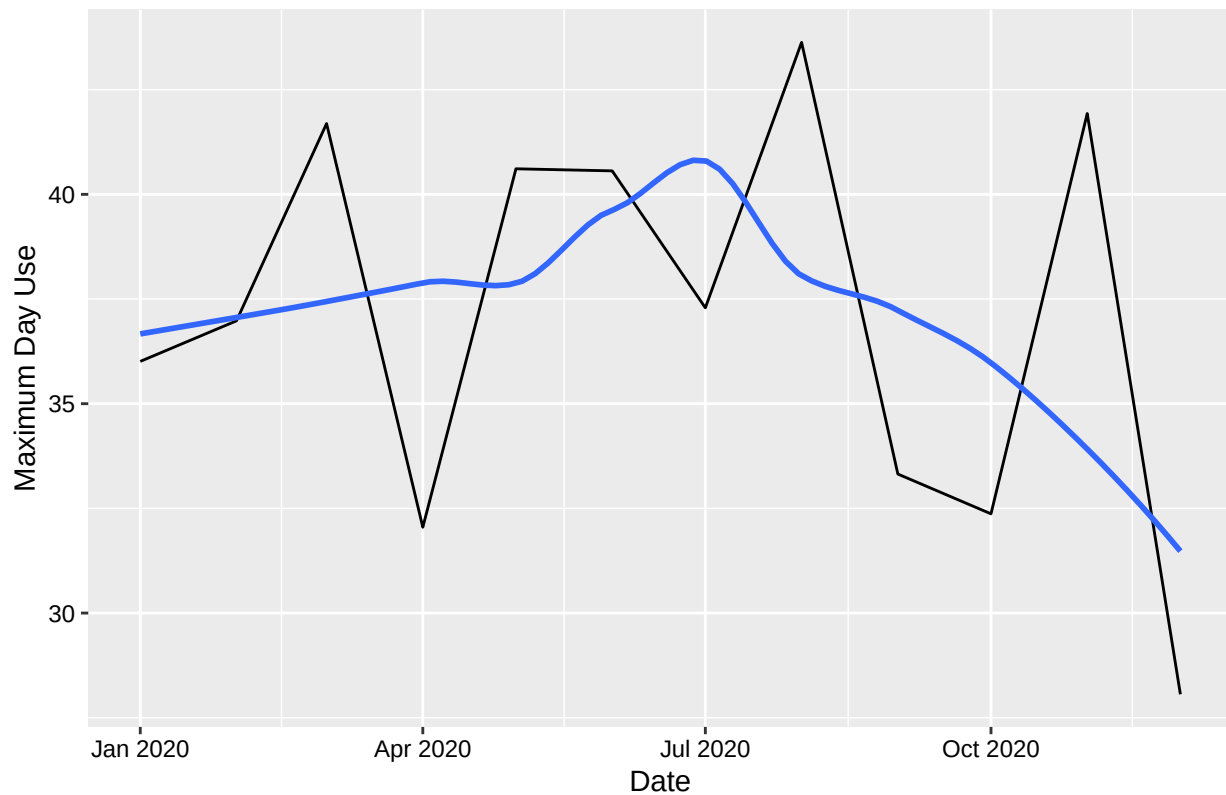
```

```

## 'geom_smooth()' using formula = 'y ~ x'

```

Maximum Day Use (MGD) in 2020



- Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

#8

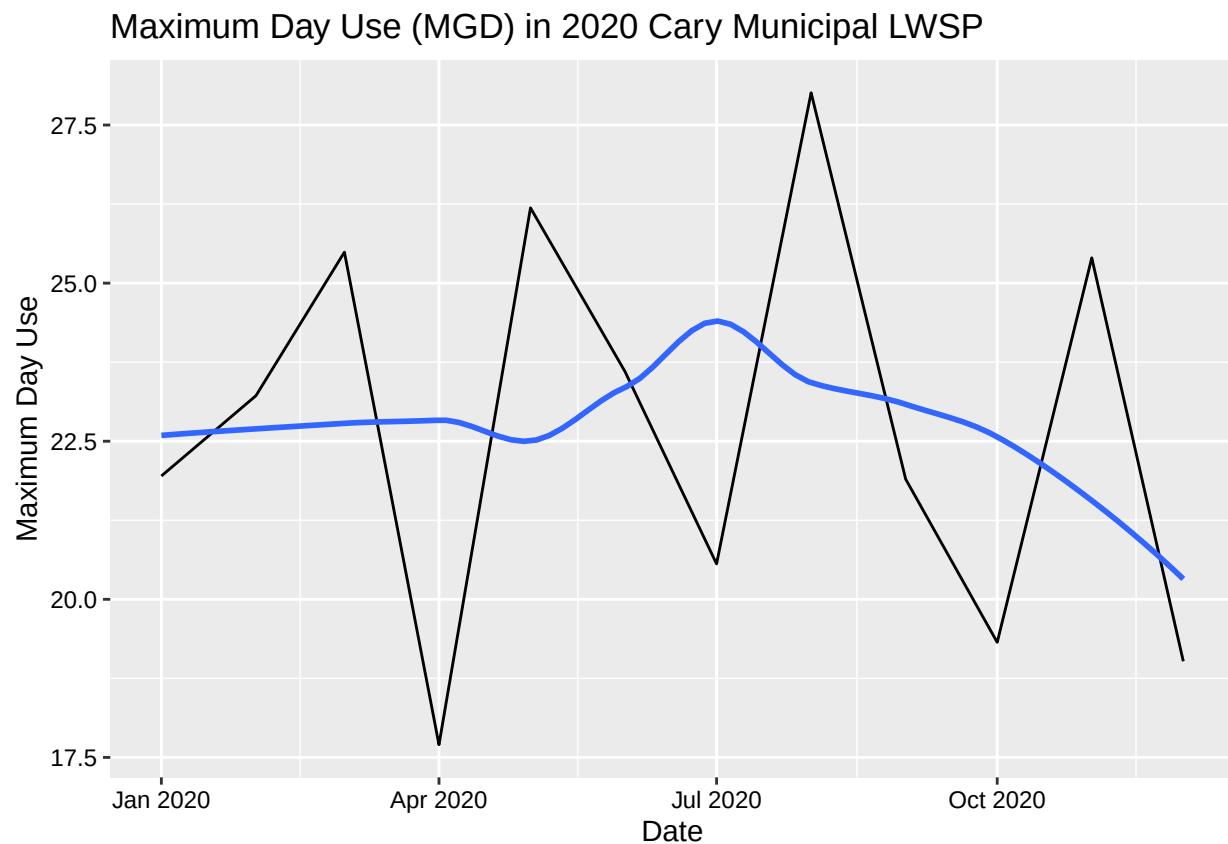
```
df_cary <- scrape.it(2020, "03-92-020")
str(df_cary)
```

```
## 'data.frame': 12 obs. of 7 variables:
## $ Month : chr "Jan" "Feb" "Mar" "Apr" ...
## $ Year : num 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ Maximum_Day_Use: num 21.9 23.2 25.5 17.7 26.2 ...
## $ Water_System : chr "Cary" "Cary" "Cary" "Cary" ...
## $ PWSID : chr "03-92-020" "03-92-020" "03-92-020" "03-92-020" ...
## $ Ownership : chr "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Date : Date, format: "2020-01-01" "2020-02-01" ...
```

```
ggplot(df_cary, aes(x=Date, y=Maximum_Day_Use)) +
  geom_line() +
  geom_smooth(method="loess", se=FALSE) +
  labs(
    title = paste(
      "Maximum Day Use (MGD) in 2020",
      subtitle = "Cary Municipal LWSP"
    ),
```

```
y="Maximum Day Use",
x="Date"
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
str(df_cary)
```

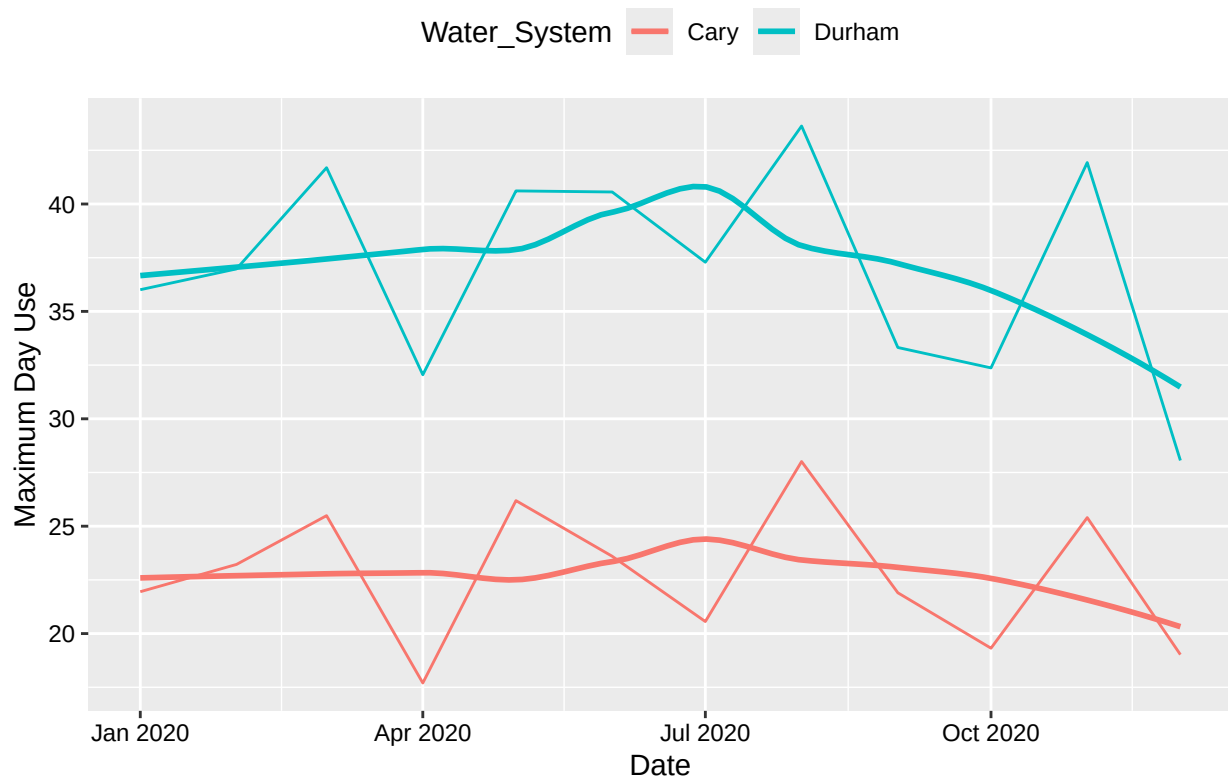
```
## 'data.frame': 12 obs. of 7 variables:
## $ Month : chr "Jan" "Feb" "Mar" "Apr" ...
## $ Year : num 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ Maximum_Day_Use: num 21.9 23.2 25.5 17.7 26.2 ...
## $ Water_System : chr "Cary" "Cary" "Cary" "Cary" ...
## $ PWSID : chr "03-92-020" "03-92-020" "03-92-020" "03-92-020" ...
## $ Ownership : chr "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Date : Date, format: "2020-01-01" "2020-02-01" ...
```

```
combine_plots <- bind_rows(df_2020, df_cary)
ggplot(combine_plots, aes(x=Date, y=Maximum_Day_Use, color = Water_System)) +
  geom_line() +
  geom_smooth(method="loess", se=FALSE) +
  labs(
    title = "Maximum Daily Withdrawals between Durham and Cary",
    y="Maximum Day Use",
```

```
x="Date"
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Maximum Daily Withdrawals between Durham and Cary



9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9
library(purrr)
library(dplyr)
library(ggplot2)

cary_id <- "03-92-020"
years <- c(2018, 2019, 2020, 2021, 2022, 2023)

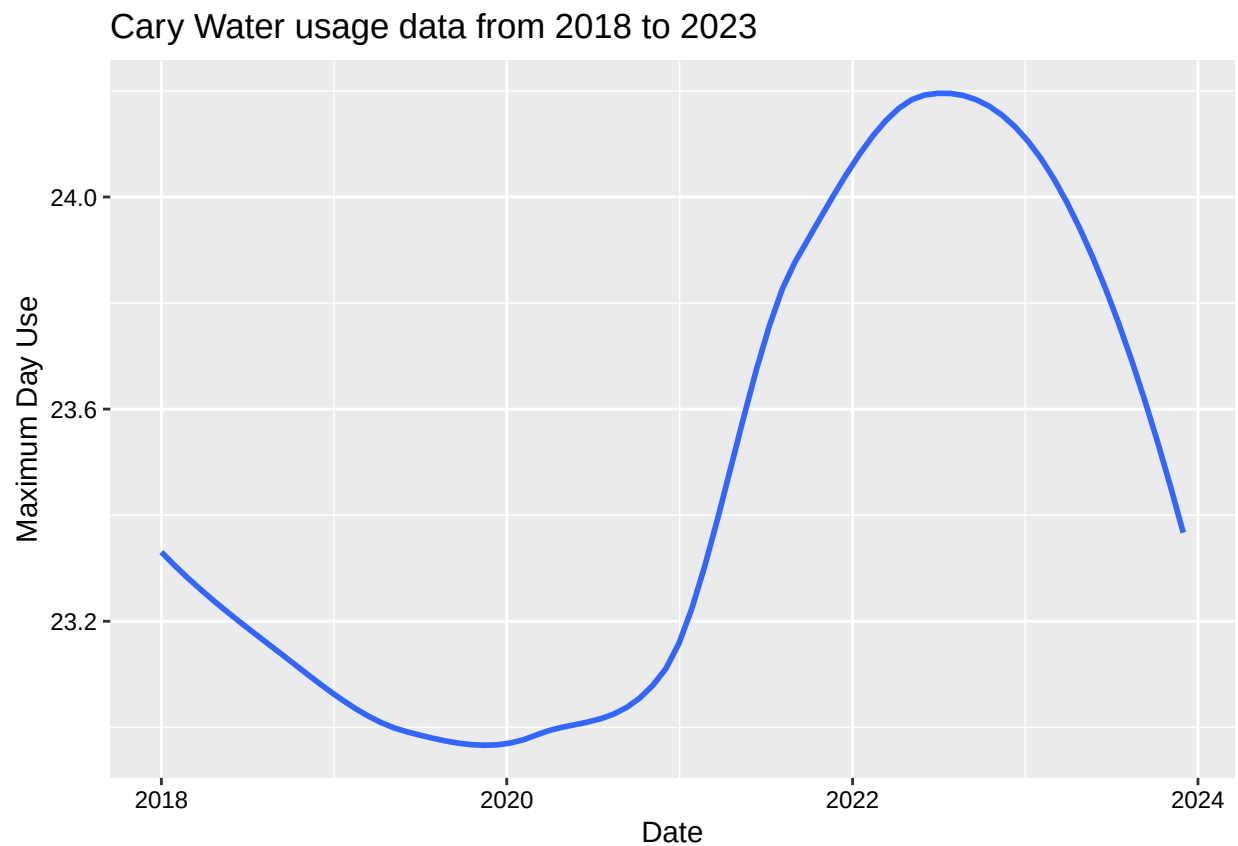
the_df <- map2(years, cary_id, scrape.it)
df_combine <- bind_rows(the_df)
```



```
ggplot(df_combine,aes(x=Date,y=Maximum_Day_Use,color=Date)) +
  #geom_line() +
  geom_smooth(method="loess",se=FALSE) +
  labs(title = "Cary Water usage data from 2018 to 2023",
        y="Maximum Day Use",
        x="Date")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: There is a clear trend in water usage in Cary where 2020 being the lowest > and 2022 being the highest. There is a drop after 2022. >